(3.3) Extrema of real valued functions

Definition $f: U \subset \mathbb{R}^n \to \mathbb{R}$

$x_0 \in U$ is a local minimum if there is a neibourhood V of x_0 such that for any $x \in V$ we have

$$f(x) \leq f(x_0), \; x \in V$$

Point $\vec{x}_0$ is critical point if

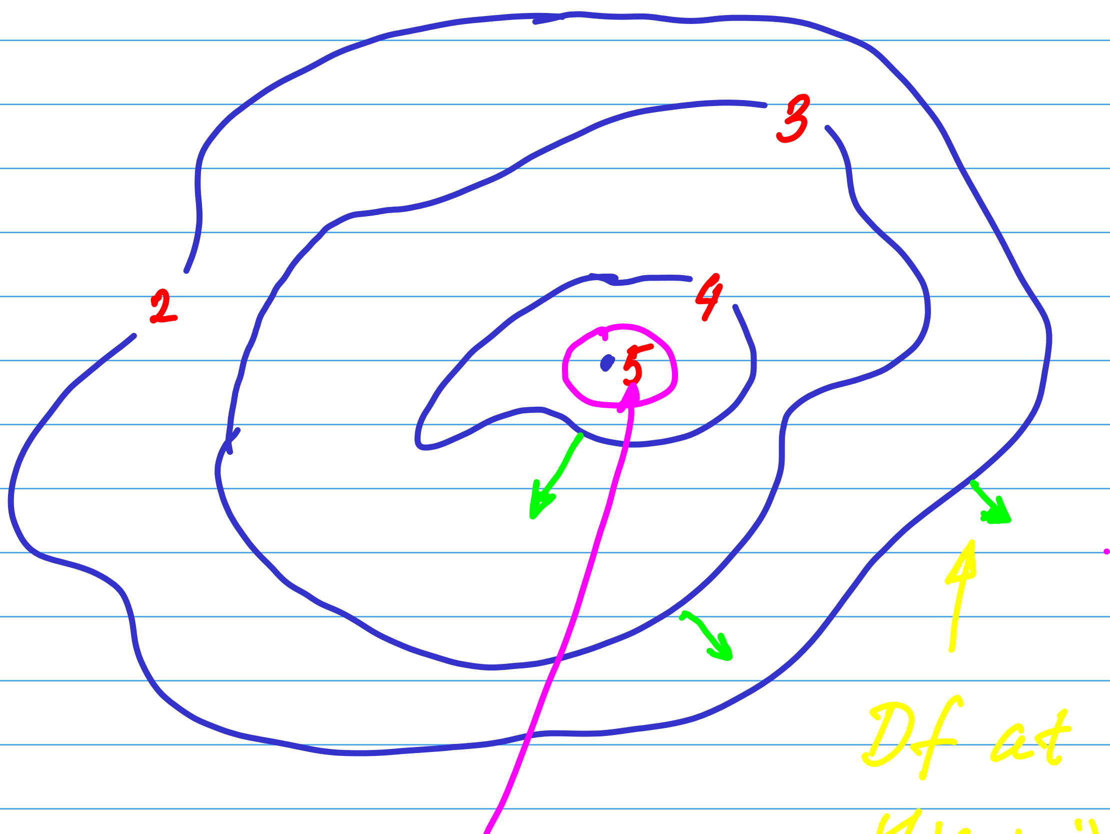
$$Df(x_0) = 0 \iff \frac{\partial f}{\partial x_i}(x_0) = 0.$$

Local minima and maxima are called $\boxed{\text{extrema}}$

Thm x_0 is a local extrema

$\Longrightarrow$ $Df(x_0) = 0$ (that is x_0 is a crit point).

Picture



here there is no prefered direction for a fast descent, hence $Df = 0$

Example $f(x,y)=x^2+y^2$

What are extrema of this funct?

$$\frac{\partial f}{\partial x}=2x \qquad \frac{\partial f}{\partial y}=2y$$

Hence $Df(x,y)=0 \iff x=y=0$

That is we have only one crit. point $(x,y)=(0,0)$ and this point is a local minimum.

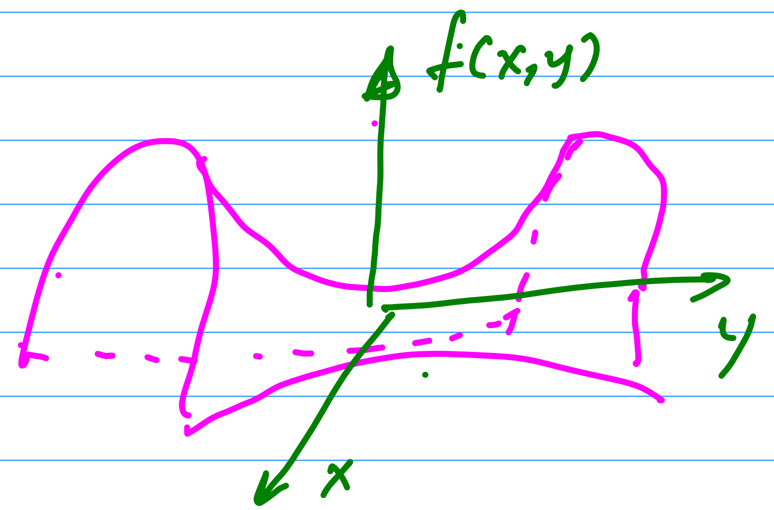
Example $f(x,y)=x^2-y^2$

What are local minima and maxima?

$$\frac{\partial f}{\partial x}=2x \qquad \frac{\partial f}{\partial y}=-2y \Rightarrow$$

$$Df(x,y)=\vec{0} \iff (x,y)=(0,0)$$

but the point $(x,y)=(0,0)$
is neither maximum or minim.



The graph of the function looks like a saddle.

Example $f(x,y) = x^2y + y^2x.$
What are local max and min?

$$\frac{\partial f}{\partial x} = 2xy + y^2 \qquad \frac{\partial f}{\partial y} = 2xy + x^2$$

$$(*)\begin{cases} 2xy + y^2 = 0 \\ 2xy + x^2 = 0 \end{cases}$$

equations for the crit points.

$$\Downarrow$$

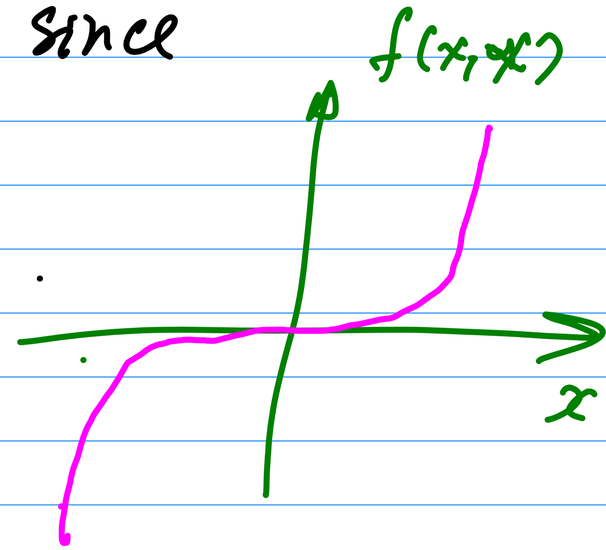
$$x^2 - y^2 = 0 \iff (x-y)(x+y) = 0$$

(1) $x = y \implies 3x^2 = 0$ (2) $x = -y$

$\implies 3y^2 = 0$

Thus the only crit point is $(x,y)=(0,0)$. But it is not max or min since

$$f(x,x) = 3x^3$$



Second derivative test.

$$f(\vec{x}_0+\vec{h}) = f(\vec{x}_0) + Hf(x_0)(h) +$$
$$R_2(\vec{x}_0,\vec{h}) \text{ if } x_0 \text{ is a crit point.}$$

Def $Hf(x_0)(\vec{h}) = \frac{1}{2}\sum \frac{\partial^2 f}{\partial x_i \partial x_j}(x_0)\, h_i h_j$

H is positive definite if

$Hf(x_0)(\vec{h}) > 0$ for any $\vec{h} \neq 0$.

It is negative definite if

$Hf(x_0)(\vec{h}) < 0$ for any $\vec{h} \neq 0$.